

lab1

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Part1

#1.

Except of itself(org.onosproject.openflow), followings apps will also be activated :

1. org.onosproject.optical-model
2. org.onosproject.hostprovider
3. org.onosproject.lldpprovider
4. org.onosproject.openflow-base

```
app active org.onosproject.openflow
```

```
reki@root > apps -a -s 20:14:06
reki@root > app activate org.onosproject.openflow 20:14:10
Activated org.onosproject.openflow
reki@root > apps -a -s 20:14:23
* 12 org.onosproject.optical-model 2.7.0 Optical Network Model
* 38 org.onosproject.hostprovider 2.7.0 Host Location Provider
* 39 org.onosproject.lldpprovider 2.7.0 LLDP Link Provider
* 40 org.onosproject.openflow-base 2.7.0 OpenFlow Base Provider
* 41 org.onosproject.openflow 2.7.0 OpenFlow Provider Suite
reki@root > 20:14:31
```

#2.

If the application Reactive Forwarding was not activated, H1 will not ping H2 successfully. Since no flows are installed on the data-plane, which forwards the traffic appropriately. Reactive Forwarding can install forwarding flows on demand, and can be activated by the command `app activate org.onosproject.fwd` or via ONOS GUI.

#3.

After activating `org.onosproject.openflow` , two TCP connections appear, which are by port 6653 and 6633. Then we can find out that **port 6653** is the TCP port that the controller listens to the OpenFlow connection request from the switch. (6633 is for the old version)

```

reki@SDN-NFV:~/SDN/Lab1$ netstat -nlpt
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.1:631          0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:5005         0.0.0.0:*               LISTEN      76309/java
tcp        0      0 0.0.0.0:6654          0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:6655          0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.53:53         0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:6656          0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:22            0.0.0.0:*               LISTEN      -
tcp6       0      0 :::631                :::*                   LISTEN      -
tcp6       0      0 :::8181                :::*                   LISTEN      76309/java
tcp6       0      0 :::8101                :::*                   LISTEN      76309/java
tcp6       0      0 127.0.0.1:37255       :::*                   LISTEN      76309/java
tcp6       0      0 :::38549               :::*                   LISTEN      3825/bazel(onos)
tcp6       0      0 :::1099                :::*                   LISTEN      76309/java
tcp6       0      0 :::9876                :::*                   LISTEN      76309/java
tcp6       0      0 :::22                  :::*                   LISTEN      -
tcp6       0      0 :::45645               :::*                   LISTEN      76309/java
reki@SDN-NFV:~/SDN/Lab1$

```

```

reki@SDN-NFV:~/SDN/Lab1$ netstat -nlpt
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.1:631          0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:5005         0.0.0.0:*               LISTEN      76309/java
tcp        0      0 0.0.0.0:6654          0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:6655          0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.53:53         0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:6656          0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:22            0.0.0.0:*               LISTEN      -
tcp6       0      0 :::631                :::*                   LISTEN      -
tcp6       0      0 :::8181                :::*                   LISTEN      76309/java
tcp6       0      0 :::8101                :::*                   LISTEN      76309/java
tcp6       0      0 127.0.0.1:37255       :::*                   LISTEN      76309/java
tcp6       0      0 :::6653                :::*                   LISTEN      76309/java
tcp6       0      0 :::6633                :::*                   LISTEN      76309/java
tcp6       0      0 :::1:38549             :::*                   LISTEN      3825/bazel(onos)
tcp6       0      0 :::1099                :::*                   LISTEN      76309/java
tcp6       0      0 :::9876                :::*                   LISTEN      76309/java
tcp6       0      0 :::22                  :::*                   LISTEN      -
tcp6       0      0 :::45645               :::*                   LISTEN      76309/java
reki@SDN-NFV:~/SDN/Lab1$

```

#4.

By deactivating

1. org.onosproject.optical-model
2. org.onosproject.hostprovider
4. org.onosproject.ldapprovider
4. org.onosproject.openflow-base

one by one, and we can know when deactivating **org.onosproject.openflow-base**, the connection will break, then it is the app that enables the controller to listen on the TCP port.

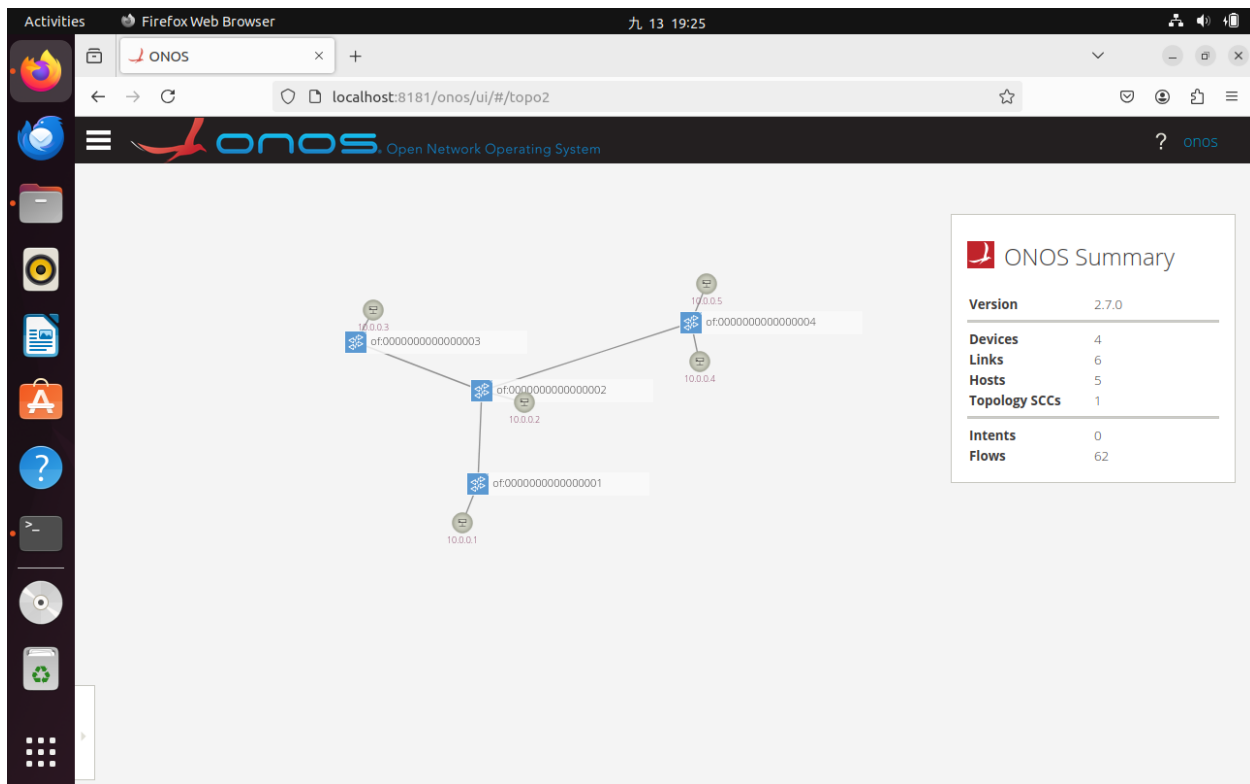
Part2

Add hosts, switches, and links by Python as the topology show

```
# Add hosts
h1 = self.addHost( 'h1' )
h2 = self.addHost( 'h2' )
h3 = self.addHost( 'h3' )
h4 = self.addHost( 'h4' )
h5 = self.addHost( 'h5' )

# Add switches
s1 = self.addSwitch( 's1' )
s2 = self.addSwitch( 's2' )
s3 = self.addSwitch( 's3' )
s4 = self.addSwitch( 's4' )

# Add links
self.addLink( h1, s1 )
self.addLink( h2, s2 )
self.addLink( h3, s3 )
self.addLink( s1, s2 )
self.addLink( s2, s3 )
self.addLink( h4, s4 )
self.addLink( h5, s4 )
self.addLink( s2, s4 )
```



Part3

Set the IP address of each host and netmask while setting hosts

```
# Add hosts
```

```
h1 = self.addHost( 'h1', ip='192.168.0.1/27' )
h2 = self.addHost( 'h2', ip='192.168.0.2/27' )
h3 = self.addHost( 'h3', ip='192.168.0.3/27' )
h4 = self.addHost( 'h4', ip='192.168.0.4/27' )
h5 = self.addHost( 'h5', ip='192.168.0.5/27' )
```

```
mininet> dump
<Host h1: h1-eth0:192.168.0.1 pid=80401>
<Host h2: h2-eth0:192.168.0.2 pid=80403>
<Host h3: h3-eth0:192.168.0.3 pid=80405>
<Host h4: h4-eth0:192.168.0.4 pid=80407>
<Host h5: h5-eth0:192.168.0.5 pid=80409>
<OVSSwitch{'protocols': 'OpenFlow14'} s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None pid=80414>
<OVSSwitch{'protocols': 'OpenFlow14'} s2: lo:127.0.0.1,s2-eth1:None,s2-eth2:None,s2-eth3:None,s2-eth4:None pid=80417>
<OVSSwitch{'protocols': 'OpenFlow14'} s3: lo:127.0.0.1,s3-eth1:None,s3-eth2:None pid=80420>
<OVSSwitch{'protocols': 'OpenFlow14'} s4: lo:127.0.0.1,s4-eth1:None,s4-eth2:None,s4-eth3:None pid=80423>
<RemoteController{'ip': '127.0.0.1', 'port': 6653} c0: 127.0.0.1:6653 pid=80395>
```

```
mininet> h1 ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.1 netmask 255.255.255.224 broadcast 192.168.0.31
    inet6 fe80::1c03:16ff:feef:30e5 prefixlen 64 scopeid 0x20<link>
    ether 1e:03:16:ef:30:e5 txqueuelen 1000 (Ethernet)
    RX packets 76 bytes 8417 (8.4 KB)
    RX errors 0 dropped 32 overruns 0 frame 0
    TX packets 25 bytes 1846 (1.8 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
mininet> h2 ifconfig
h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.2 netmask 255.255.255.224 broadcast 192.168.0.31
    inet6 fe80::84e5:a2ff:fe75:8217 prefixlen 64 scopeid 0x20<link>
    ether 86:e5:a2:75:82:17 txqueuelen 1000 (Ethernet)
    RX packets 79 bytes 8908 (8.9 KB)
    RX errors 0 dropped 34 overruns 0 frame 0
    TX packets 25 bytes 1846 (1.8 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
mininet> h3 ifconfig
h3-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.3 netmask 255.255.255.224 broadcast 192.168.0.31
    inet6 fe80::6c15:5fff:fe47:50d prefixlen 64 scopeid 0x20<link>
    ether 6e:15:5f:47:05:0d txqueuelen 1000 (Ethernet)
    RX packets 85 bytes 9742 (9.7 KB)
    RX errors 0 dropped 40 overruns 0 frame 0
    TX packets 25 bytes 1846 (1.8 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
mininet> h4 ifconfig
h4-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.4 netmask 255.255.255.224 broadcast 192.168.0.31
    inet6 fe80::ccdd:7bff:feb9:6360 prefixlen 64 scopeid 0x20<link>
    ether ce:dd:7b:b9:63:60 txqueuelen 1000 (Ethernet)
    RX packets 92 bytes 10683 (10.6 KB)
    RX errors 0 dropped 46 overruns 0 frame 0
    TX packets 26 bytes 1916 (1.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
mininet> h5 ifconfig
h5-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.5 netmask 255.255.255.224 broadcast 192.168.0.31
    inet6 fe80::3809:ecff:fe49:d462 prefixlen 64 scopeid 0x20<link>
    ether 3a:09:ec:49:d4:62 txqueuelen 1000 (Ethernet)
    RX packets 97 bytes 11309 (11.3 KB)
    RX errors 0 dropped 50 overruns 0 frame 0
    TX packets 26 bytes 1916 (1.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Part4

```

reki@root > app deactivate org.onosproject.lldpprovider
Deactivated org.onosproject.lldpprovider
reki@root > apps -a -s
* 25 org.onosproject.drivers          2.7.0    Default Drivers
* 38 org.onosproject.hostprovider     2.7.0    Host Location Provider
* 69 org.onosproject.gui2             2.7.0    ONOS GUI2
* 110 org.onosproject.fwd             2.7.0    Reactive Forwarding

```

在測試Part1 #4時，原本我不管 deactivate 哪一個 app 連線都會斷掉，後來我用 `apps -a -s` 確認後發現在我 deactivate 任何一個時，其他的 app 也會自動被關掉，因此要手動將它們 activate 回來才有辦法進行測試。

這次的實驗中我學會如何用 Python Script 簡單的建立一個網路，並認識了 ONOS、mininet 等工具，製作自己的網路及將其以圖表示出來，同時我也學會利用 netstat 觀察連線狀況，了解連線是透過甚麼方式建立的。

References

Basic ONOS Tutorial | ONOS Wiki

<https://wiki.onosproject.org/display/ONOS/Basic+ONOS+Tutorial>

Port 6653 | speedguide.net

<https://www.speedguide.net/port.php?port=6653>

Port 6633 | speedguide.net

<https://www.speedguide.net/port.php?port=6633>

軟體定義網路 - OpenFlow Ports 介紹 | hackmd

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